

AGENCYOS

PHASE 4 - TASK 2

Full UI Design + UI Prototype

IMPLEMENTATION PLAN + MASTER CHECKLIST + TESTING & ACCEPTANCE SPECIFICATION

Status	LOCKED PHASE 4 IMPLEMENTATION BASELINE
Previous Task	Task 1 - UI Finalizing / Phase 3
Current Task	Task 2 - Complete UI Design + UI Prototype
Primary Input	Client-approved theme + colors + screen/content baseline + approved scope
Primary Output	Client-approved complete UI + client-approved prototype + structured Phase 5 handoff
Finance Gate	Phase 4 completion → mandatory M2 = 20%; Phase 5 waits for Admin-verified payment
Canonical UI Source	Exact approved Figma UI version / truthful equivalent when Figma write automation is unavailable
QA Principle	Designer/Prototype Agent cannot self-certify; QA retest evidence is required

This document converts the locked Phase 4 business flow into implementation order, domain/state contracts, engineering acceptance criteria, master checklists and a detailed test catalogue.

1. Source Basis & Precedence

Source	Purpose in this document
Phase 4 Master Flow / Implementation Plan	Task 2 sequence, PM communication, review order, M2 trigger and Phase 5 gate
UI Design & Prototype Workflow Specification	Screen inventory, coverage, responsive/accessibility, UI versioning and prototype rules
QA Testing & Production Readiness	Independent QA, defect lifecycle, regression/security/performance/compatibility principles
Finance, Invoice, Payment & Milestone System	Milestone/payment separation and Admin verification
Automation, Events & Job System	Events, queues/jobs, retries and idempotency
Security, Permissions, Audit & Data Governance	RBAC/RLS, tenant isolation, secrets and audit
Database & Domain Model	Canonical/versioned design/prototype records
Development Roadmap & Definition of Done	Repository-first oaa analysis and evidence-based completion

Precedence: latest locked Phase 4 decisions govern Task 2 behavior. Master specs supply supporting architecture unless they conflict with a newer explicit Phase 4 rule.

2. Objective, Entry & Exit

2.1 Objective

Convert the locked Phase 3 visual direction into the complete high-fidelity UI, independently verify it, obtain Admin/client approval, build a versioned interactive prototype from that exact UI, independently verify the prototype, obtain final client approval, complete Task 2, and trigger the M2 payment milestone.

2.2 Entry Conditions

- ☐ Phase 3 / Task 1 completed.
- ☐ Client-approved theme and color combination explicitly recorded.
- ☐ Final Phase 3 screen list and screen-content baseline exist.
- ☐ Approved scope/requirement version is known.
- ☐ Project Planning context is available.
- ☐ Phase 3 Figma/design source is uniquely identifiable.
- ☐ PM Task 1 completion / Task 2 start communication can be recorded/sent.

2.3 Exit Conditions

- ☐ Complete UI coverage validated against approved scope.
- ☐ Design QA passes the exact UI version.
- ☐ Admin approves the exact UI version.
- ☐ Client explicitly approves the exact UI version.
- ☐ Approved UI is locked as the prototype source.
- ☐ Prototype is built from the exact locked UI version.
- ☐ Prototype QA passes the exact build.
- ☐ Admin approves the exact prototype build.
- ☐ Client explicitly approves the final prototype build.
- ☐ Phase 4 Definition of Done passes and Phase4Completed is recorded.
- ☐ Finance M2 = 20% is triggered.
- ☐ Phase 5 remains blocked until Admin verifies M2 payment.

3. Roles & Boundaries

Role	Owns	Must not do
PM Agent	Task announcements, client clarification, review coordination, exact-version sharing, revisions, completion communication	Design UI, self-approve QA/Admin/client gates, verify payment
UI/Figma Designer Agent	Complete high-fidelity UI, required states/variants, revisions, exact design-version artifacts	Invent scope, self-approve, bypass PM/Admin/client order
QA Agent / QA capability	Independent design QA and prototype QA, defects, evidence, retest	Grant Admin/client approval or change scope
UI Prototype Agent	Interactive prototype/build from exact locked UI, routing/interactions/mock data/states	Build from stale/unapproved UI, self-approve QA
Orchestrator / Coordination	Routing, state, dependencies, jobs, retries, idempotency, evidence	Replace human approval authority
Finance Agent	M2 invoice/payment workflow after Phase 4 completion	Verify payable money without Admin
Admin	Human UI/prototype approval; M2 payment verification	N/A - human authority
Client	Human UI/prototype review and explicit confirmation	N/A - human authority

Admin and Client are humans. Orchestrator/Coordination is workflow infrastructure. Finance activates after Task 2 completion.

4. Locked End-to-End Flow

TASK 1 COMPLETE → PM TASK 2 ANNOUNCEMENT → LOAD PHASE 3 BASELINE → UI COVERAGE CHECK → COMPLETE UI → DESIGN QA → FIX/RETEST → ADMIN UI REVIEW → CLIENT UI REVIEW → REVISION LOOP → CLIENT UI APPROVAL → UI LOCK → PROTOTYPE BUILD → PROTOTYPE QA → FIX/RETEST → ADMIN PROTOTYPE REVIEW → CLIENT PROTOTYPE REVIEW → REVISION LOOP → CLIENT FINAL APPROVAL → TASK 2 COMPLETE →

AgencyOS state is authoritative. WhatsApp/client messages reflect workflow state; retries must not duplicate business transitions.

5. Implementation Workstreams

WS-01 Phase 4 Bootstrap

- ☐ Create/extend Phase4Workspace linked to project, Phase 3 baseline and active scope.
- ☐ Validate Phase 3 completion before Task2Started.
- ☐ Make phase/task start idempotent and auditable.

WS-02 Coverage Engine

- ☐ Map Requirement → Feature → Screen → Interaction → State → Prototype → QA.
- ☐ Flag missing required screens, orphaned/out-of-scope screens, duplicate screens and role-state gaps.
- ☐ Prevent readiness if mandatory coverage is incomplete.

WS-03 Full High-Fidelity UI

- ☐ Build every required approved screen from locked Phase 3 design direction.
- ☐ Include applicable loading, empty, validation, error, success, permission, offline and overflow states.
- ☐ Reuse tokens/components/assets; do not re-decide theme/colors.
- ☐ Persist exact design/Figma file/page/node/version or truthful equivalent metadata.

WS-04 Design QA

- ☐ Create independent DesignQAReport for exact UI version.
- ☐ Record structured defects with screen/requirement/evidence/severity.
- ☐ Designer fixes; QA retests; only QA verification closes defect.

WS-05 Admin UI Review

- ☐ Generate Admin package: exact UI version, scope version, coverage, QA evidence, known limitations, revision history.
- ☐ Support CONFIRM/EDIT.

- ☐ EDIT routes Designer → QA → Admin again.

WS-06 Client UI Review

- ☐ PM shares only Admin-approved exact UI version.
- ☐ Capture exact client feedback/evidence and classify correction/revision/clarification/scope change/design-direction change.
- ☐ Version revisions; preserve old client-shared versions.
- ☐ Approx. 2-3 client revision rounds then human escalation.

WS-07 UI Lock

- ☐ Require explicit client approval.
- ☐ Lock approved UI as read-only baseline and prototype source.
- ☐ Later changes create a new version rather than mutating approved baseline.

WS-08 Prototype Build

- ☐ Create prototype from exact locked UI.
- ☐ Support approved type: web preview, Android APK, configured iOS preview/distribution, or other agreed type.
- ☐ Implement critical flows, meaningful navigation, representative/mock data and important states.
- ☐ Store build ID, source UI version, artifact and limitations.

WS-09 Prototype QA

- ☐ Verify critical routes/buttons/screens, approved visual consistency, overflow, important states and exact source link.
- ☐ Defects route Prototype Agent → fix → QA retest.
- ☐ Compilation/build success alone is not QA PASS.

WS-10 Prototype Review

- ☐ Only QA-passed build reaches Admin.
- ☐ Admin EDIT routes Prototype Agent → QA → Admin.
- ☐ PM shares only Admin-approved build to client.
- ☐ Client feedback references exact build; final explicit approval locks build.

WS-11 Completion & Finance

- ☐ Validate Phase 4 DoD.
- ☐ Record Task 2 completion / Phase4Completed.
- ☐ Trigger mandatory M2 = 20%.
- ☐ Payment proof remains pending verification.
- ☐ Only Admin-verified M2 opens Phase 5.

6. Required Domain & Data Contracts

Entity	Core contract
Phase4Workspace	project_id, organization_id, status, source_phase3_version, scope_version, current_ui_version_id, current_prototype_build_id, revision counters, blockers, timestamps
UIScreen	screen_definition_id, ui_version_id, role, purpose, design/Figma node ref, required states, responsive refs, status
UIVersion	project_id, source_phase3_version, scope_version, screen set, token set, status, QA/Admin/client refs, lock metadata
UICoverageRecord	requirement/feature, screen, interaction/state, prototype route/build, QA coverage/result
DesignQAResult	ui_version_id, checks, defects, evidence, reviewer/agent, result, timestamps
UIRevision	from_version, to_version, origin QA/ADMIN/CLIENT, request/evidence, changed screens, QA/Admin/client results
PrototypeBuild	build ID/version, project_id, source_ui_version_id, type/platform, artifact/url, environment, status, limitations
PrototypeQAResult	build_id, checks, defects, evidence, result, retest refs
PrototypeRevision	from_build, to_build, origin, request, fix summary, QA/Admin/client results
PMCommunication	project/phase/task/event, template_version, rendered_message, channel, delivery state, external ref
Phase4Handoff	approved UI/prototype refs, screen inventory, tokens, flows, assets, QA/Admin/client evidence, revisions
AuditEvent	organization/project_actor_action_entity/version_before/after_evidence/correlation_timestamp

Reuse existing canonical AgencyOS entities where equivalent structures already exist; do not create duplicate parallel domain models.

7. State Machines

7.1 Task 2 / Phase 4

NOT_STARTED → UI_DESIGN → UI_QA → ADMIN_UI_REVIEW → CLIENT_UI_REVIEW → UI_LOCKED → PROTOTYPE_BUILD → PROTOTYPE_QA → ADMIN_PROTOTYPE_REVIEW → CLIENT_PROTOTYPE_REVIEW → COMPLETED → M2_PAYMENT_PENDING → M2_VERIFIED → PHASE5_READY

7.2 UI Version

DRAFT → QA_REVIEW → QA_CHANGES_REQUIRED / QA_PASS → ADMIN_REVIEW → ADMIN_EDIT / ADMIN_APPROVED → CLIENT_REVIEW → CLIENT_CHANGE / CLIENT_APPROVED → LOCKED

7.3 Prototype Build

PLANNED → BUILDING → BUILD_READY → QA_REVIEW → QA_CHANGES_REQUIRED / QA_PASS → ADMIN_REVIEW → ADMIN_EDIT / ADMIN_APPROVED → CLIENT_REVIEW → CLIENT_CHANGE / CLIENT_APPROVED → LOCKED

7.4 M2 Gate

M2_NOT_DUE → M2_DUE → INVOICED → PAYMENT_PENDING → PAYMENT_SUBMITTED → PENDING_VERIFICATION

Illegal transitions must be rejected server-side. Frontend visibility is not a workflow/security boundary.

8. Core Events & Reliability

Event	Primary effect
Phase3Completed	Evaluate Task 2 eligibility
Task2Started / Phase4Started	Initialize Phase 4 workspace
UIDesignReady	Create Design QA job
UIDesignQACHangesRequired	Designer revision
UIDesignQAPassed	Create Admin UI review
AdminUIEditRequested / AdminUIApproved	Revision or client-share eligibility
UISharedToClient / ClientUIChangeRequested / ClientUIApproved	Client UI loop and UI lock
PrototypeBuildReady / PrototypeQAPassed	Prototype QA/Admin workflow
AdminPrototypeEditRequested / AdminPrototypeApproved	Revision or client-share eligibility
PrototypeSharedToClient / ClientPrototypeChangeRequested / ClientPrototypeApproved	Client prototype loop
Phase4Completed	Trigger Finance M2
M2PaymentVerified	Open Phase 5 financial gate
Task3Started	Start Phase 5 eligibility

- ☐ Every async job has explicit identity/status and persisted result.
- ☐ Duplicate events do not create duplicate UI versions, builds, PM messages or invoices.
- ☐ Phase4Completed → M2 trigger is idempotent.
- ☐ Jobs preserve organization/project scope.
- ☐ Failure exposes reason, retry count and next action in Admin operations.

9. PM Communication Implementation

Message	Trigger	Evidence
Task 1 Complete / Task 2 Start	Valid Phase 3 completion	task/event ref, rendered message, delivery status
Complete UI Ready	Design QA PASS + Admin UI approval	exact UI version and share refs
UI Changes Received	Client allowed feedback	source message + revision record
UI Approved / Prototype Start	Explicit client UI approval	approval evidence + locked UI version
Prototype Ready	Prototype QA PASS + Admin approval	exact build ID/artifact
Prototype Changes Received	Client allowed feedback	source feedback + revision record
Task 2 Complete	Final prototype approval + DoD	completion event + final build
M2 Verified / Task 3 Start	Admin-verified M2	finance verification + Task3 event

Templates remain configurable. Persist actual rendered/sent content and template version so history does not change later.

10. Admin Panel Requirements

Surface	Must expose
Phase 4 Overview	Task state, owners, blockers, current UI/prototype versions, revision counts, next gate
Phase 3 Baseline	selected theme/colors, screen/content baseline, source design/Figma refs
UI Coverage	requirement/feature/screen/state/prototype/QA traceability
Full UI	all UI versions, screen inventory, exact refs, lock state
Design QA	defects, evidence, fixes, retest and PASS
Admin UI Review	CONFIRM/EDIT decisions and history

Surface	Must expose
Client UI Review	exact versions shared, feedback, classification, approval
Prototype Builds	build IDs, source UI version, type/platform, artifact, limitations
Prototype QA	defects, retest and PASS
Admin Prototype Review	CONFIRM/EDIT decisions
Client Prototype Review	exact build shared, feedback, final approval
PM Communication	task announcements/review messages, delivery state, failures/retries
Finance M2	invoice, payment submitted, pending verification, Admin verification, Phase 5 gate
Activity / Audit	versions, actors, events, approvals, retries and state transitions

11. Security, Permissions & Governance

- ☐ Designer can write design artifacts but cannot grant QA/Admin/client approval.
- ☐ QA can record pass/fail evidence but cannot impersonate Admin/client decisions.
- ☐ Prototype Agent can create builds but cannot self-approve them.
- ☐ PM can coordinate/share but cannot perform Admin approval or payment verification.
- ☐ Finance manages M2 workflow but Admin verifies payable money.
- ☐ All records are organization/project scoped and protected server-side.
- ☐ Figma/provider/build credentials remain server-side.
- ☐ Client artifact links use appropriate access controls.
- ☐ Sensitive transitions and approvals create audit events.
- ☐ Cross-project and cross-tenant access is denied by default and tested.

12. Cost & Duplicate-Work Controls

- ☐ Reuse Phase 3 theme/colors/tokens/components/screen definitions.
- ☐ Do not regenerate the complete UI for a small targeted revision.
- ☐ Retry does not duplicate expensive design/prototype generation.
- ☐ Use deterministic code for states, revision counts, coverage and permissions.
- ☐ Use cheaper suitable models for simple work; stronger reasoning only where needed.
- ☐ Keep AI provider abstraction; do not hardwire Phase 4 to one provider.
- ☐ Record real usage/cost where available; do not invent zero cost when unknown.
- ☐ Revision-limit escalation prevents uncontrolled client iteration cost.

13. Technical Implementation Plan

P4-01 Repository & Gap Discovery

Map existing Phase 3 handoff, design domain, QA, prototype tooling, PM communication, Admin Panel, Finance M2, events/jobs, security and tests.

Acceptance: Gap matrix exists; no duplicate architecture before discovery.

P4-02 Phase 4 Domain & State

Implement/extend workspace, versioning, revisions, state guards and audit hooks.

Acceptance: Illegal transitions rejected; project/tenant scope enforced.

P4-03 Baseline Loader

Load locked theme/colors/design ref/screen baseline/scope/project plan/assets.

Acceptance: Stale/missing baseline blocks start; known context reused.

P4-04 PM Communication

Implement Task 2 start/review/completion/Task 3 messages with outbox/retry/history.

Acceptance: Message retry cannot duplicate business transitions.

P4-05 Coverage Engine

Build requirement→screen→state→prototype→QA traceability.

Acceptance: Missing/orphan/duplicate coverage detectable.

P4-06 Full UI Workflow

Create all required screens/states/variants using locked Phase 3 direction.

Acceptance: Exact version identifiable; no silent scope expansion.

P4-07 Design QA

Independent QA report, defects, evidence, fix/retest.

Acceptance: Defects close only after QA verification.

P4-08 Admin UI Review

Admin package + Confirm/Edit + re-review loop.

Acceptance: Only QA PASS reaches Admin.

P4-09 Client UI Review

Exact-version share, feedback classification, revisions, scope route, revision limit.

Acceptance: Exact client-shared version preserved.

P4-10 UI Lock

Explicit client approval → immutable/uniquely identified baseline.

Acceptance: Prototype rejects unapproved UI.

P4-11 Prototype Build

Versioned prototype linked to exact UI with artifact metadata.

Acceptance: Mock/test data separate from production.

P4-12 Prototype QA

Independent critical-flow/build verification with defect retest.

Acceptance: Build success alone cannot pass.

P4-13 Admin Prototype Review

Admin package + Confirm/Edit + re-review loop.

Acceptance: Only QA PASS reaches Admin.

P4-14 Client Prototype Review

Exact-build share, revisions/scope classification and final approval.

Acceptance: Final approval references exact build.

P4-15 Handoff Package

Package approved UI/prototype/screens/flows/tokens/assets/evidence.

Acceptance: Phase 5 can consume without re-deciding UI.

P4-16 Finance M2

Phase4Completed → mandatory 20% M2 → Admin verification gate.

Acceptance: No duplicate M2; payment proof alone cannot open Phase 5.

P4-17 Admin Visibility

Expose full Task 2 operational history and next action.

Acceptance: Admin does not need SQL/logs to understand status.

P4-18 Security Hardening

Validate RBAC/RLS, scoping, secrets and artifact access.

Acceptance: Cross-tenant/project access denied.

P4-19 Reliability

Idempotency, retries, failure visibility and recovery.

Acceptance: Safe replay and resumable failures.

P4-20 Full Verification

Run full test matrix + regression and fix P0/P1.

Acceptance: DoD evidence complete; no unresolved P0/P1.

14. Testing Strategy - Required Layers

Layer	Phase 4 focus
Unit	state guards, revision counters, coverage logic, permission predicates, M2 trigger conditions
Database	schema/constraints, version links, project/tenant scoping, unique/idempotency rules, migrations
API / Server	validation, exact-version actions, authorization, errors and idempotency
Integration	Phase 3 handoff, design/Figma refs, QA↔Designer, communication, prototype artifact, Finance M2
Agent	structured inputs/outputs, boundaries, timeout/retry and no gate bypass
Event / Job	producer/consumer, replay, retry, failure recovery and audit
Admin UI	routes, tables, drawers, filters, approval actions, loading/empty/error/read-only/permission states
Design QA	screen/state/responsive coverage, visual consistency and version linkage
Prototype QA	routes/interactions, important states, source version, artifact integrity and secrets
Security	authz/RBAC/RLS, tenant/project isolation, artifact access and privilege escalation
E2E	happy path, UI revision, prototype revision, M2 and Phase 5 gate
Recovery	provider timeout, worker crash, message/build failure and event replay
Regression	Phase 1-3 and shared Finance/communication behavior

15. Test Evidence Rules

- ☐ Every test run identifies project/fixture, exact UI version/build, environment and timestamp.
- ☐ Use synthetic/test data separate from production.
- ☐ Automated results are machine-readable where tooling supports it.
- ☐ Manual visual/exploratory checks record useful evidence for important defects.
- ☐ Defects close only after QA retest evidence.
- ☐ Do not claim formal accessibility compliance unless that scope actually runs.
- ☐ Do not invent universal performance thresholds; use project/configured targets.
- ☐ Regression failures remain visible and block completion according to severity/policy.

16. Detailed Phase 4 Test Catalogue

ID	Category	Test	Expected result
P4-T001	Entry	Start Task 2 with valid Phase 3 completion	Workspace starts once; exact source baseline stored
P4-T002	Entry	Start Task 2 with Phase 3 incomplete	Transition blocked with missing condition
P4-T003	Entry	Replay Task2Started	No duplicate workspace/message/job
P4-T004	Baseline	Load locked theme/colors/screen baseline	Exact Phase 3 version reused
P4-T005	Baseline	Design/Figma source missing	Explicit blocker/manual-assisted state; no fake success
P4-T006	Coverage	Requirement missing required UI screen	Coverage fails/flags gap
P4-T007	Coverage	Screen has no approved scope mapping	Flag potential out-of-scope design
P4-T008	Coverage	Duplicate screen	Flag duplicate; do not count as complete
P4-T009	Coverage	Role state missing	Coverage remains incomplete
P4-T010	UI	Create full UI version	Linked to exact scope/Phase 3 baseline
P4-T011	UI	Required loading/empty/error/success states	Present or explicit N/A reason
P4-T012	UI	Required responsive/platform variant	Variant exists and is linked
P4-T013	Scope	Designer attempts new out-of-scope feature	Rejected/routed to scope-change workflow
P4-T014	Design QA	QA finds missing state	Defect created; QA PASS withheld
P4-T015	Design QA	Designer marks defect fixed	Still unverified until QA retest
P4-T016	Design QA	QA retest passes	Defect verified/closed
P4-T017	Admin UI	Send UI to Admin before QA PASS	Blocked
P4-T018	Admin UI	Admin CONFIRM	UI becomes client-share eligible
P4-T019	Admin UI	Admin EDIT	Designer → QA → Admin again
P4-T020	Client UI	PM shares approved exact UI version	Share record stores exact version/artifacts
P4-T021	Client UI	Attempt share of non-approved version	Blocked
P4-T022	Client UI	Client visual revision within allowance	New version + controlled review loop
P4-T023	Scope	Client requests new functionality	Routed to Change Request
P4-T024	Revision	Revision limit reached	Automatic loop stops; human escalation
P4-T025	UI Lock	Client approves exact UI version	Version locks as prototype source
P4-T026	UI Lock	Attempt to mutate locked version	Rejected; new revision required
P4-T027	Prototype	Build from locked UI	Build stores exact source UI version
P4-T028	Prototype	Build from unapproved UI	Rejected
P4-T029	Prototype	Use mock data	Separated from production data
P4-T030	Security	Prototype includes production secret	Security FAIL; build blocked
P4-T031	Prototype QA	Broken critical route	QA FAIL + defect
P4-T032	Prototype QA	Fix and retest route	QA verifies revised build
P4-T033	Prototype QA	Major overflow on target viewport	Defect; severity-based block
P4-T034	Prototype QA	Build/source UI mismatch	QA FAIL
P4-T035	Admin Prototype	Send to Admin before QA PASS	Blocked
P4-T036	Admin Prototype	Admin EDIT	Prototype Agent → QA → Admin again
P4-T037	Client Prototype	PM shares exact approved build	Share stores build ID/artifact
P4-T038	Client Prototype	Client requests allowed correction	New build/revision + QA/Admin repeat
P4-T039	Scope	Client requests new feature in prototype	Scope-change path
P4-T040	Completion	Client explicitly approves final prototype	Eligible for DoD evaluation
P4-T041	Completion	Try Phase4Completed with missing approval evidence	Blocked
P4-T042	Finance	Phase4Completed triggers M2	Exactly one 20% milestone path

ID	Category	Test	Expected result
P4-T043	Finance	Replay Phase4Completed	No duplicate M2 invoice
P4-T044	Finance	Payment proof submitted	Remains pending verification
P4-T045	Finance	Unauthorized payment verification	Denied
P4-T046	Finance	Admin verifies payment	M2 verified; gate may open
P4-T047	Phase 5 Gate	Start Phase 5 before M2 verified	Blocked
P4-T048	Phase 5 Gate	Start after M2 verified	Eligible; Task3 message can run
P4-T049	Communication	Task 2 start message delivery fails	Business state preserved; retryable
P4-T050	Communication	Message retry	No duplicate phase transition
P4-T051	Security	Cross-project UI access	Denied
P4-T052	Security	Cross-tenant prototype access	Denied
P4-T053	Security	Designer tries Admin approval	Denied
P4-T054	Security	PM tries payment verification	Denied
P4-T055	Event	Duplicate UIDesignReady	Single QA business outcome
P4-T056	Recovery	QA worker crashes	Safe retry from persisted state
P4-T057	Recovery	Prototype build worker fails	Failure visible; retry safe
P4-T058	Recovery	Figma/provider unavailable	Explicit blocker/manual-assisted state
P4-T059	Admin UI	No Phase 4 data	Useful empty state, no fake metrics
P4-T060	Admin UI	Long names/feedback	No broken layout; detail remains accessible

17. Mandatory End-to-End Scenarios

17.1 Happy Path

- ☐ Phase 3 completed with exact baseline locked.
- ☐ PM Task 2 start communication recorded/sent.
- ☐ Coverage validation passes.
- ☐ Designer creates complete UI.
- ☐ Design QA passes exact UI version.
- ☐ Admin approves UI.
- ☐ PM shares exact UI; client explicitly approves.
- ☐ UI locks.
- ☐ Prototype built from exact locked UI.
- ☐ Prototype QA passes exact build.
- ☐ Admin approves prototype.
- ☐ PM shares exact build; client explicitly approves.
- ☐ Task 2 completes; Phase4Completed fires.
- ☐ Finance issues M2 20%.
- ☐ Payment proof remains pending verification.
- ☐ Admin verifies M2.
- ☐ PM Task 3 start runs; Phase 5 becomes eligible.

17.2 UI Revision Path

- ☐ QA defect → Designer fix → QA retest.
- ☐ Admin EDIT → Designer → QA → Admin.
- ☐ Client included revision → Designer → QA → Admin → PM re-share.
- ☐ Client approves only the new exact UI version; that version locks.

17.3 Prototype Revision Path

- ☐ Prototype QA defect → Prototype Agent fix → QA retest.
- ☐ Admin EDIT → Prototype Agent → QA → Admin.
- ☐ Client allowed correction → new build → QA/Admin → PM re-share.
- ☐ Older client-shared builds remain historical.

17.4 Scope-Change Path

- ☐ Client request introduces new feature/module/role/major flow.
- ☐ Feedback is classified as possible scope change.
- ☐ Normal revision does not silently implement it.
- ☐ Requirements/Scope Change workflow receives the request.

17.5 Failure / Recovery Path

- ☐ Message delivery failure is retryable without duplicate business transition.
- ☐ Design/Figma/provider failure creates a truthful blocker/manual-assisted state.
- ☐ Prototype worker failure is safely retryable.
- ☐ Critical event replay remains idempotent.

18. Manual / External Dependencies

- ☐ If Figma write/create is unsupported, expose assisted/manual action instead of fake automatic design creation.
- ☐ If platform-specific prototype distribution requires external credentials/account approval, keep build ready and show the dependency.
- ☐ Admin and client approvals remain intentional human governance steps.
- ☐ M2 payment verification remains intentional Admin/human control.
- ☐ Manual blockers store owner, reason, expected evidence and resume condition.

19. Phase 4 Definition of Done

- ☐ Valid Phase 3 → Phase 4 handoff verified.
- ☐ Task 2 start communication and delivery tracking work.
- ☐ Locked Phase 3 baseline reused without re-deciding theme/color.
- ☐ Coverage matrix is complete.
- ☐ All required screens/states/platform variants represented.
- ☐ Design QA independently passes exact UI version.
- ☐ Admin UI approval works.
- ☐ Client UI revision/final explicit approval works.
- ☐ Approved UI is locked; historical versions preserved.
- ☐ Prototype built from exact locked UI.
- ☐ Prototype QA independently passes exact build.
- ☐ Admin prototype approval works.
- ☐ Client prototype revision/final explicit approval works.
- ☐ Final build locked; historical builds preserved.
- ☐ Phase 4 handoff package complete.
- ☐ Task 2 completion communication works.
- ☐ Phase4Completed triggers mandatory M2 20%.
- ☐ Payment proof cannot auto-open Phase 5.
- ☐ Only Admin-verified M2 opens Phase 5.
- ☐ Admin Panel shows versions, QA, approvals, PM messages, M2 and next action.
- ☐ Idempotency/retry/recovery pass.
- ☐ Security and tenant/project isolation pass.
- ☐ Regression confirms Phase 1-3 shared functionality still works.
- ☐ No unresolved Phase 4 P0/P1 defects.

20. Master Implementation Checklist

A. Entry & Baseline

- ☐ Phase 3 completed
- ☐ theme/colors locked
- ☐ design/Figma version
- ☐ scope version
- ☐ screen/content baseline
- ☐ project plan
- ☐ Task2Started idempotency

B. PM Communication

- ☐ Task2 start
- ☐ UI ready
- ☐ UI changes
- ☐ UI approved/prototype start
- ☐ prototype ready
- ☐ prototype changes
- ☐ Task2 complete
- ☐ M2 verified/Task3 start
- ☐ delivery history

C. Coverage

- ☐ requirements
- ☐ features
- ☐ roles

- ☐ screens
- ☐ states
- ☐ responsive/platform
- ☐ prototype
- ☐ QA
- ☐ orphan/duplicate checks

D. Full UI

- ☐ all screens
- ☐ states
- ☐ tokens/components
- ☐ assets
- ☐ variants
- ☐ accessibility basics
- ☐ exact versioning

E. Design QA

- ☐ QA intake
- ☐ coverage checks
- ☐ defects
- ☐ evidence
- ☐ Designer fix
- ☐ QA retest
- ☐ PASS

F. UI Approval

- ☐ Admin package
- ☐ Confirm/Edit
- ☐ client share
- ☐ feedback classification
- ☐ revisions
- ☐ scope routing
- ☐ revision limit
- ☐ client approval
- ☐ UI lock

G. Prototype

- ☐ exact UI source
- ☐ build type
- ☐ routes
- ☐ interactions
- ☐ mock data
- ☐ validation
- ☐ states
- ☐ artifact/build ID
- ☐ limitations

H. Prototype QA

- ☐ routes
- ☐ buttons
- ☐ overflow
- ☐ states
- ☐ source match
- ☐ secrets

- ☐ defects
- ☐ fix/retest
- ☐ PASS

I. Prototype Approval

- ☐ Admin package
- ☐ Confirm/Edit
- ☐ client share
- ☐ feedback
- ☐ revisions
- ☐ client final approval
- ☐ build lock

J. Handoff

- ☐ UI version
- ☐ prototype build
- ☐ screens
- ☐ tokens
- ☐ flows
- ☐ assets
- ☐ QA/Admin/client evidence
- ☐ limitations

K. Finance

- ☐ Phase4Completed
- ☐ M2 20%
- ☐ invoice
- ☐ delivery
- ☐ payment
- ☐ pending verification
- ☐ Admin verification
- ☐ Phase5 gate

L. Admin Panel

- ☐ overview
- ☐ Phase3 baseline
- ☐ coverage
- ☐ UI versions
- ☐ Design QA
- ☐ reviews
- ☐ prototype builds
- ☐ Prototype QA
- ☐ PM messages
- ☐ Finance M2
- ☐ audit

M. Security & Reliability

- ☐ RBAC
- ☐ RLS/project scope
- ☐ secrets
- ☐ artifact access
- ☐ idempotency
- ☐ retry
- ☐ failed jobs

☐ audit

N. Testing

☐ unit

☐ database

☐ API

☐ integration

☐ agent

☐ event/job

☐ Admin UI

☐ design QA

☐ prototype QA

☐ security

☐ E2E

☐ negative

☐ recovery

☐ regression

21. Evidence & Traceability Checklist

- ☐ Every Phase 4 requirement has a source requirement/section reference.
- ☐ Existing implementation is classified before rewriting.
- ☐ Changed files/services/migrations map to requirement IDs.
- ☐ Every state-changing action has authorization and audit evidence.
- ☐ Every async event/job has idempotency/retry evidence.
- ☐ Every QA PASS references exact UI version/build.
- ☐ Every Admin/client decision references exact artifact version.
- ☐ Every client share stores the exact version/build sent.
- ☐ Every revision preserves from-version/to-version and origin.
- ☐ Phase4Completed has objective evidence, not only a UI status/button.
- ☐ M2 is traceable to Phase4Completed.
- ☐ Phase 5 gate test proves payment proof alone is insufficient.
- ☐ Manual/external blockers are explicit and resumable.

22. Final Locked Summary

PHASE 3 COMPLETE → TASK 2 START → LOCKED BASELINE → COVERAGE VALIDATION → COMPLETE UI →
DESIGN QA → ADMIN → CLIENT → REVISIONS → CLIENT UI APPROVAL → LOCK UI → PROTOTYPE BUILD →
PROTOTYPE QA → ADMIN → CLIENT → REVISIONS → CLIENT FINAL APPROVAL → PHASE 4 COMPLETE → M2

Completion rule: Phase 4 is not complete because screens exist or a prototype compiles. It is complete only when exact versions are independently QA-verified, Admin/client-approved, locked, traceable, handed forward, and the M2 financial gate is triggered correctly.